Started on	Sunday, 24 July 2022, 5:56 PM
State	Finished
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Time taken	7 secs
Marks	0.00/69.00
Grade	**0.00** out of 100.00

Question **1**

Not answered

Marked out of 1.00

Flag question

Which of the following is best choice to treat alcohol withdrawal in a man with severe liver damage?

Select one:

- Naltrexone
- Carbamazepine
- Chlordiazepoxide
- Lorazepam
- Diazepam

Explanation:

When managing alcohol withdrawal, a short-acting benzodiazepine such as oxazepam or lorazepam may be used in individuals with impaired liver function.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 459.

The correct answer is: Lorazepam

Question **2**

Not answered

Marked out of 1.00

Flag question

A patient drives to work at 6:30 AM and wants night sedation. Based on half life, which of the following is the best?

Select one:

- Lorazepam
- Zopiclone
- Zolpidem
- Temazepam
- Diazepam

Explanation:

Zolpidem is rapidly and well absorbed after oral administration. Zolpidem reaches peak

plasma concentrations in 1.6 hours and has a half-life of 2.6 hours. The rapid metabolism and lack of active metabolites of zolpidem avoid the accumulation of plasma concentrations compared with the use of benzodiazepines such as diazepam.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 679.

The correct answer is: Zolpidem

Question **3**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs is excreted unchanged in urine?

Select one:

- Quetiapine
- Gabapentin
- Valproate
- Diazepam
- Morphine

Explanation:

Gabapentin, sulpiride, amisulpride, and lithium all undergo no or minimal hepatic metabolism.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 724.

The correct answer is: Gabapentin

Question **4**

Not answered

Marked out of 1.00

Flag question

Which of the following psychotropic will be affected most by contact with moist air?

Select one:

- Olanzapine
- Gabapentin
- Carbamazepine
- Topiramate
- Sodium valproate

Deliquescent salts such as valproate absorb moisture from the air

The correct answer is: Sodium valproate

Question **5**

Not answered

Marked out of 1.00

A psychotropic drug is observed to undergo phase 2 metabolism directly without undergoing phase 1 reactions. Which one of the following fits with the above description best?

Select one:

- Lithium

Flag question

- Fluoxetine
- Chlordiazepoxide
- Diazepam
- Lorazepam

Explanation:

Most benzodiazepines are oxidised first by CYP3A4 and CYP2C19, often to active metabolites. These metabolites may then be hydroxylated to another active metabolite. However, lorazepam, oxazepam, and temazepam are metabolised by non-CYP450 pathways. They are conjugated directly by glucuronidation and are excreted.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 670.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 780.

The correct answer is: Lorazepam

Question **6**

Not answered

Marked out of 1.00

Flag question

Which one among the following statements is least likely to be correct with respect to the tissue distribution of drugs?

Select one:

- Drugs with high affinity for tissues such as fat have high volume of distribution
- The drug distribution in plasma to various tissues depends on plasma protein binding
- The drug distribution in plasma to various tissues depends on tissue permeability
- Distribution is uniform across the body tissues
- Tissue distribution leads to a fall in plasma concentration

Drugs are not evenly distributed in the body. There are three main compartments through which a drug may be distributed. (1) The Vascular compartment (2) The extra cellular compartment (3) The intracellular compartment.Drugs that are large molecules and those which bind tightly to plasma proteins remain in the vascular compartment; drugs with low lipid solubility are distributed in the vascular plus the extracellular compartment whilst lipid soluble drugs are distributed through the three compartments. Drugs such as Phenytoin, diazepam, chlorpromazine and amitriptyline are all highly bound to plasma proteins. They are very lipophilic and once absorbed pass readily into peripheral fat stores; from which they are gradually released long after the last dose has been administered.

The correct answer is: Distribution is uniform across the body tissues

Question **7**

Not answered

Marked out of 1.00

Which of the following medications must be used with caution when fluvoxamine is prescribed?

Select one:

- Theophylline

Flag question

- Naproxen
- Amlodipine
- Salbutamol
- Paracetamol

Explanation:

Fluvoxamine is a potent inhibitor of CYP1A2 which can result in increased theophylline serum levels.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 308.

The correct answer is: Theophylline

Question **8**

Not answered

Marked out of 1.00

Flag question

Which of the following acts on presynaptic receptors as the main mechanism of action?

Select one:

- Donepezil
- Diazepam
- Phenelzine
- Propranolol
- Clonidine

Explanation:

Clonidine and guanfacine are presynaptic alpha-2-receptor agonists. They inhibit sympathetic outflow and cause vasodilation of the blood vessels. The agonist effects of clonidine and guanfacine on pre-synaptic alpha-2-adrenergic receptors in the sympathetic nuclei of the brain result in a decrease in the amount of norepinephrine released from the presynaptic nerve terminals. This decrease serves to reset the body's sympathetic tone at a lower level and decrease arousal.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 694.

The correct answer is: Clonidine

Question **9**

Not answered

Marked out of 1.00

Flag question

Hepatic metabolism may involve various chemical reactions. Which one of the following is not a common chemical activity seen in hepatic metabolism?

Select one:

- oxidation
- reduction
- dealkylation
- hydroxylation

- polymerisation

Hepatic metabolism may involve non-synthetic reactions that involve oxidation, reduction, hydroxylation, dealkylation, demethylation, acetylation; methylation may produce active compounds (Phase 1). The metabolism also involves synthetic reactions such as conjugation usually with glucuronic acid and sulphation that produce inactive water-soluble compounds. (Phase 2)

The correct answer is: polymerisation

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs has its plasma levels reduced even after regular administration of the same dose for 2 weeks?

Select one:

- Lithium
- Clozapine
- Chlormethiazole
- Valproate
- Carbamazepine

Explanation:

Carbamazepine induces its own metabolism. Time to steady state is dependent on this autoinduction and is usually at least 2 weeks.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 835.

The correct answer is: Carbamazepine

Question **11**

Not answered

Marked out of 1.00

Flag question

Enzyme autoinhibition is most likely to be seen with which of the following antidepressants?

Select one:

- Sertraline
- Duloxetine
- Fluoxetine
- Paroxetine
- Citalopram

Explanation:

Paroxetine is a potent CYP450 2D6 inhibitor. It inhibits its own metabolism, so dosing is not linear. Plasma concentrations can double when oral doses increase by 50%. Plasma concentrations can increase by 2-7 fold when oral doses are doubled.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021.

pp. 596-599.

The correct answer is: Paroxetine

Question **12**

Not answered

Marked out of 1.00

Flag question

Under which of the following conditions interactions with drugs that alter protein binding becomes important?

Select one:

- Obesity
- Children
- Lung disease
- High altitude living
- Renal disease

Protein binding interactions become relevant in the presence of a renal disease where proteinuria can occur.

The correct answer is: Renal disease

Question **13**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs has a higher therapeutic index than the others listed?

Select one:

- Lithium
- Phenytoin
- Type 1 anti-arrhythmics
- Paracetamol
- Tricyclics

Explanation:

The therapeutic index is any of several indices relating the clinical effectiveness of a drug to its safety factor, one being the therapeutic ratio of the median lethal dosage to the median effective dosage (LD-50/ED-50). Other therapeutic indices include the ratio of the minimum toxic dosage to the minimum effective dosage, and the difference between the minimum effective dosage and the minimum toxic dosage. Drugs available over the counter usually have a better (higher) therapeutic index (or else they won't be sold without a prescription). This does not mean that they are absolutely safe.

Ref: Colman AM. Oxford Dictionary of Psychology, 4th Edition. 2015.

The correct answer is: Paracetamol

Question **14**

Not answered

A 44-year-old man is on warfarin for thromboprophylaxis. He returns from a holiday to mainland Europe where he obtained a preparation for treating his depression. The GP is concerned about his INR and organizes an urgent blood test. What is the likely drug taken?

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Flag question

Select one:

- St John's Wort
- Duloxetine
- Carbamazepine
- Citalopram
- Venlafaxine

Explanation:

St. John's wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2c39, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: St John's Wort

Question **15**

Not answered

Marked out of 1.00

Flag question

Which of the following is least likely to be a pharmacokinetic property of the tricyclic antidepressants?

Select one:

- Extensive hepatic breakdown
- Their activity may delay their own absorption
- Lipid solubility
- Metabolism being reduced by smoking
- Extensive protein binding in the plasma

Explanation:

Smoking facilitates metabolism and generally reduces plasma levels of tricyclic antidepressants by 25-50%. Tricyclics readily cross lipid barriers such as blood-brain barrier and placenta. They are extensively bound to plasma proteins. Due to anticholinergic effects, they may reduce their own absorption.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 858.

The correct answer is: Metabolism being reduced by smoking

Question **16**

Not answered

Marked out of

A drug X is used to treat acute behavioural disturbances. It can be given orally or intravenously. 15mg of the oral drug is needed for an acutely disturbed man of weight 70kgs while 10mg is sufficient if given intravenously. What is the bioavailability of IV administration

1.00

Flag question

of 10mg?

Select one:

- 10.5mg
- 10mg
- 5mg
- 1.5mg
- 15mg

When a drug is administered intravenously the availability of the drug is 100%.

The correct answer is: 10mg

Question **17**

Not answered

Marked out of 1.00

Flag question

Steady-state plasma concentration is usually achieved after

Select one:

- 1-2 half-lives
- 2-3 half-lives
- 10-20 half-lives
- 4-5 half-lives
- 9-10 half-lives

Explanation:

Steady state concentration is achieved after repeated doses, which lead to an equilibrium between absorption and elimination. It usually takes 4-5 drug half-lives to reach steady-state levels.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 833.

The correct answer is: 4-5 half-lives

Question **18**

Not answered

Marked out of 1.00

Flag question

Some drugs such as fluoxetine move from first order to zero order kinetics in supratherapeutic doses. What happens to their t1/2 in such cases?

Select one:

- t1/2 doubles
- t1/2 becomes independent of dose
- t1/2 becomes dependent on dose
- t1/2 quadruples
- t1/2 halves

In very high supratherapeutic doses, saturation of enzymes can happen for drugs such as fluoxetine, wherein first order elimination switches to become zero order. In zero-order kinetic state, a constant amount, rather than a constant fraction, of the drug is cleared per unit time. This means that irrespective of the amount of drug in plasma or dose of drug administered, only a fixed unit of drug is cleared by the body. Here the concept of half-life does not hold true as 'half life' depends on the dose administered (i.e the time taken to eliminated 50% of dose, becomes dose dependent, and the concept of t1/2 loses its validity).

The correct answer is: t1/2 becomes dependent on dose

Question **19**

Not answered

Marked out of 1.00

Flag question

In a patient with severe liver disease, the antipsychotic of choice is

Select one:

- Amisulpride
- Clozapine
- Quetiapine
- Risperidone
- Olanzapine

Explanation:

Amisulpride is predominantly renally excreted, so dosage reduction should not be necessary as long as renal function is normal. Amisulpride is uncommonly associated with rises in transaminases and hepatocellular injury.

Drug group	Recommended psychotropics in hepatic impairment
Antipsychotics	**Sulpiride/amisulpiride:** no dosage reduction required if renal function is normal **Paliperidone:** if depot required
Antidepressants	**Paroxetine, sertraline, citalopram, or vortioxetine**: start at low dose; titrate slowly if required
Mood stabilisers	**Lithium:** use plasma levels to guide dosage; care needed if ascites status changes
Sedatives	**Lorazepam, oxazepam, temazepam**: as short half-life with no active metabolites; use low doses with caution, as sedative drugs used in severe disease can precipitate hepatic encephalopathy **Zopiclone**: 3.75mg with care in moderate hepatic impairment
	Adapted from Maudsley Prescribing Guidelines, 2021. p. 730.

SPMM COURSE

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 724, 730.

The correct answer is: Amisulpride

Question **20**

Which one among the following drugs can decrease lithium excretion and precipitate toxicity?

Not answered

Marked out of 1.00

Flag question

Select one:

- Diazepam
- Valproate
- Propranolol
- Haloperidol
- NSAIDs

Explanation:

NSAIDs inhibit the synthesis of renal prostaglandins, thereby reducing renal blood flow and possibly increasing renal re-absorption of sodium and therefore lithium. The magnitude of the rise is unpredictable for any given patient; case reports vary from increases of around 10% to over 400%.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p 253.

The correct answer is: NSAIDs

Question **21**

Not answered

Marked out of 1.00

Flag question

A drug X, when given at a dose of 300 units, is cleared from the body at a rate of 30 units per hour. When the dose is increased to 600 units, it is still cleared at a rate of 30 units per hour. Which of the following is true? The drug follows

Select one:

- first order kinetics
- zero order at lower doses and first order at higher doses
- first order at lower doses and zero order at higher doses
- no specific kinetics
- zero order kinetics

When the system facilitating first order clearance of drugs gets saturated, drugs follow zero-order kinetics. Here a constant amount, not a fraction, of the drug is cleared per unit time. This means that irrespective of the amount of drug in plasma or dose of drug administered, only a fixed unit of the drug is cleared by the body. As such, increasing dose might result in serious toxicity in this case.

The correct answer is: zero order kinetics

Question **22**

Not answered

Marked out of 1.00

Flag question

What is least likely to be correct concerning the pharmacology of clozapine?

Select one:

- High affinity towards D4 receptors
- Slow release from D2 receptors
- Has antimuscarinic properties

- Acts on D1 receptors
- Blocks 5HT2A receptors

Explanation:

Clozapine is an antagonist of 5-HT2A, D1, D3, and D4 receptors. It also has potent antagonism of muscarinic M1, H1, and alpha-1 adrenergic receptors. It has relatively low potency as a D2 receptor antagonist. Clinically effective doses of clozapine occupy only 40-50% of striatal D2 receptors. This is probably why clozapine has low propensity to cause EPS. It has also been postulated that clozapine binds more loosely to the D2 receptor, and because of this 'fast dissociation', more normal dopamine neurotransmission is possible.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 613.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 225.

The correct answer is: Slow release from D2 receptors

Question **23**

Not answered

Marked out of 1.00

Flag question

Bioavailability is least likely to be influenced by

Select one:

- Metabolism
- Absorption
- Distribution
- Excretion
- Gender

Bioavailability is the fraction of an ingested dose of a drug that gains access to the systemic circulation. It may be low because the drug is metabolised in the gut wall or liver before reaching the systemic circulation.

The correct answer is: Gender

Question **24**

Not answered

Marked out of 1.00

Flag question

A fatal combination with respect to potentially severe serotonin syndrome is

Select one:

- Moclobemide and paroxetine
- Paroxetine and methylphenidate
- MDMA and moclobemide
- Pethidine and tranylcypromine
- Phenelzine and amitriptyline

Explanation:

Serotonin toxicity, often referred to as serotonin syndrome, is an iatrogenic drug-induced toxidrome associated with raised intra-synaptic concentrations of serotonin (5-HT) in the CNS. Dangerous toxic reactions, including death, have been shown to result after the administration of meperidine (pethidine – an opioid analgesic drug used in anaesthesia) to patients taking an MAOI such as phenelzine or tranylcypromine.

Ref: Baldo BA, Rose MA. The anaesthetist, opioid analgesic drugs, and serotonin toxicity: a mechanistic and clinical review. British journal of anaesthesia. 2020 Jan 1;124(1):44-62.

The correct answer is: Pethidine and tranylcypromine

Question **25**

Not answered

Marked out of 1.00

Flag question

Which of the following hypnotic drugs acts for the longest duration?

Select one:

- Temazepam
- Triazolam
- Zolpidem
- Zaleplon
- Nitrazepam

Explanation:

Nitrazepam is a benzodiazepine compound with sedative properties. The half-life is on average 24 hours. In elderly patients the volume of distribution is greater, and the mean elimination half-life rises to 40 hours.

Ref: https://www.medicines.org.uk/emc/product/7145/smpc#gref

The correct answer is: Nitrazepam

Question **26**

Not answered

Marked out of 1.00

Flag question

A clear therapeutic window is noted for which of the following?

Select one:

- Nortriptyline
- Amitriptyline
- Carbamazepine
- Fluoxetine
- Clozapine

Explanation:

Although blood levels are available for many antidepressants; those for imipramine, desipramine, and nortriptyline have been best established. Imipramine and desipramine appear to have a curvilinear dose-response curve with an optimal range of 150 to 300 ng/mL. Nortriptyline appears to have a therapeutic window in the range of 50 to 150 ng/mL, usually reached by doses ranging from 50-100mg/day. Note that in clinical practice, therapeutic drug monitoring for nortriptyline is rarely used and of dubious benefit. If toxicity is suspected, ECG

is the best investigation.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 836.

The correct answer is: Nortriptyline

Question **27**

Not answered

Marked out of 1.00

Flag question

When plotting plasma concentration against time, the area under the curve after a single dose allows determination of the

Select one:

- Plasma half-life (t-1/2)
- Metabolism
- Elimination
- Distribution
- Bioavailability of the drug

The area under the curve (AUC) is proportional to the amount of drug in plasma and allows determination of fraction of dose absorbed (i.e. bioavailability of the drug). Low bioavailability usually means high first pass metabolism. T-1/2 is the time taken for the plasma concentration to fall by a half. It only applies for drugs with first order kinetics.

The correct answer is: Bioavailability of the drug

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs is excreted unchanged almost completely via the renal route?

Select one:

- Olanzapine
- Lithium
- Quetiapine
- Amisulpride
- Carbamazepine

Explanation:

95% of lithium is excreted unchanged in the urine. Amisulpride is primarily renally excreted, but only 50% is excreted unchanged in the urine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 738, 744.

The correct answer is: Lithium

Question **29**

Not answered

Advantages of buspirone over benzodiazepines when used for anxiety include all of the following except

Marked out of 1.00

Flag question

Select one:

- Buspirone causes no dependence
- Buspirone can be given once daily
- Buspirone can be withdrawn more easily
- Buspirone has no street value
- Buspirone causes no tolerance usually

Explanation:

Because of its short half-life (2-11 hours), buspirone is dosed three times daily. Buspirone does not cause discontinuation/withdrawal symptoms.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 673.

The correct answer is: Buspirone can be given once daily

Question **30**

Not answered

Marked out of 1.00

Flag question

Which of the following can be given as a loading dose in acute mania?

Select one:

- Lithium
- Clozapine
- Valproate
- Olanzapine
- Clonazepam

Explanation:

Loading doses can be used with valproate. Achieving therapeutic plasma levels of valproate rapidly using a loading dose regimen is generally well tolerated.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 259-261.

The correct answer is: Valproate

Question **31**

Not answered

Marked out of 1.00

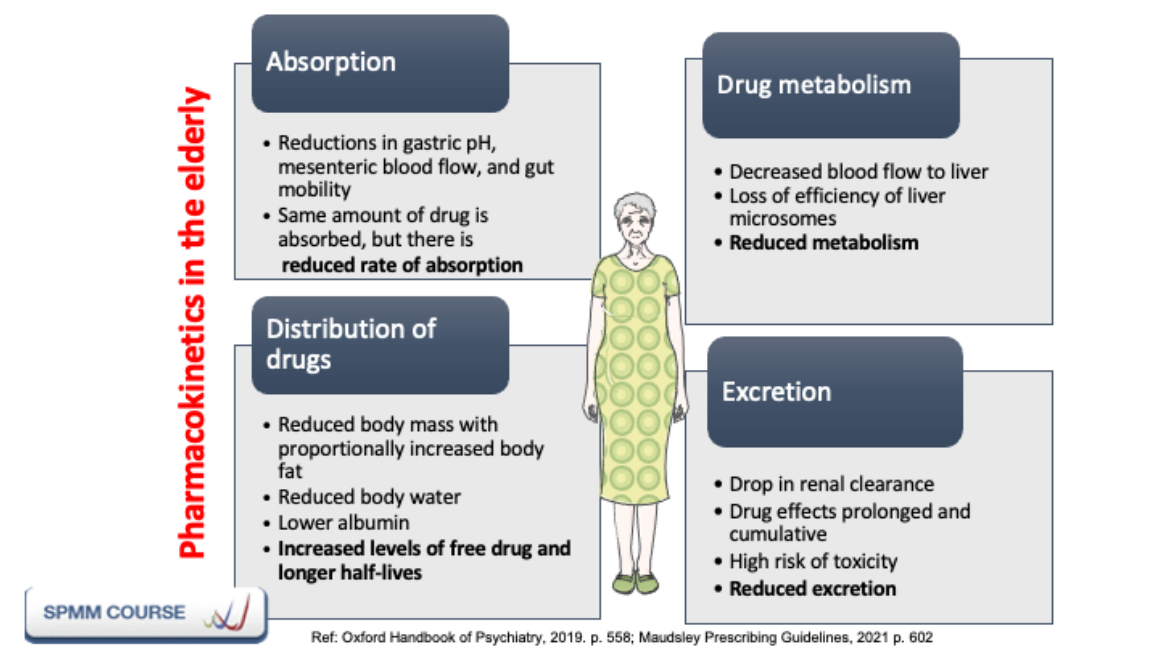
Flag question

Which of the following is not altered much when considering pharmacokinetics in the elderly?

Select one:

- Total body fat
- Renal clearance
- Plasma protein binding
- Amount of GI absorption
- Gastric pH

Explanation:



Ref: Semple D, Smyth R. Oxford handbook of psychiatry. Oxford university press; 2019 Jul 30. p. 558.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 602

The correct answer is: Amount of GI absorption

Question **32**

Not answered

Marked out of 1.00

Flag question

The drug of choice for alcohol detoxification in a patient with liver disease is

Select one:

- Chlordiazepoxide
- Diazepam
- Carbamazepine
- Chlormethiazole
- Oxazepam

Explanation:

When managing alcohol withdrawal, a short-acting benzodiazepine such as oxazepam or lorazepam may be used in individuals with impaired liver function. Oxazepam has a lesser propensity to accumulate and cause adverse reactions in the elderly or people with hepatic impairment. As it does not require hepatic oxidation, and is entirely metabolized via glucuronidation, it is a safer choice if benzodiazepines are required in those with liver failure.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 459.

The correct answer is: Oxazepam

Question **33**

Not answered

Marked out of 1.00

Flag question

Which of the following is incorrect with respect to the pharmacokinetic properties of diazepam?

Select one:

- The peak concentration is achieved within 30-90 minutes after single dose
- The protein binding ratio is 30-40%
- Bioavailability is almost complete with the oral dose
- Desmethyldiazepam is the active phase 1 metabolite
- The rate of absorption is rapid

Explanation:

All benzodiazepines except chlorazepate are completely absorbed after oral administration and reach peak serum levels within 30 minutes to 2 hours. The absorption, attainment of peak concentrations, and the onset of action are quickest for diazepam, lorazepam, alprazolam, triazolam, and estazolam. The amount of protein binding varies from 70 to 99 percent. Diazepam is oxidised to desmethyldiazepam, which, in turn, is hydroxylated to produce oxazepam. These products undergo glucuronidation to inactive metabolites.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. pp. 669-670.

The correct answer is: The protein binding ratio is 30-40%

Question **34**

Not answered

Marked out of 1.00

Flag question

The SSRI most selective for serotonin reuptake is

Select one:

- Citalopram
- Paroxetine
- Sertraline
- Duloxetine
- Fluoxetine

Explanation:

Amongst the SSRIs, citalopram and escitalopram are the most selective inhibitors of serotonin reuptake, with very little inhibition of norepinephrine or dopamine reuptake and very low affinities for histamine H1, GABA, or benzodiazepine receptors.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 626.

The correct answer is: Citalopram

Question **35**

Not answered

The active metabolite of imipramine is

Select one:

Marked out of 1.00

Flag question

- Lofepramine
- mcPP
- Desipramine
- Nortriptyline
- Amitriptyline

Explanation:

Imipramine is metabolised by CYP2C19 to desipramine, which is also a tricyclic antidepressant with distinct clinical features that differ from imipramine. Desipramine is then metabolised by CYP2D6 to the less active hydroxy-imipramine. The active metabolite of amitriptyline is nortriptyline.

Ref: Dean L. Imipramine Therapy and CYP2D6 and CYP2C19 Genotype. 2017 Mar 23. In: Pratt VM, Scott SA, Pirmohamed M, et al., editors. Medical Genetics Summaries [Internet]. Bethesda (MD): National Center for Biotechnology Information (US); Available from:https://www.ncbi.nlm.nih.gov/books/NBK425164/

The correct answer is: Desipramine

Question **36**

Not answered

Marked out of 1.00

Flag question

Regarding the pharmacokinetics of Lithium which one of the following statements in not true?

Select one:

- It passes freely through the glomerular membrane independent of serum concentration
- Lithium has a wide therapeutic range
- It is not bound to serum proteins
- Lithium renal clearance is reduced when sodium is depleted
- The half-life of lithium lies between 10 and 24 hours

Lithium has a narrow therapeutic range, i.e., when compared with other drugs, there is a narrow gap between the minimum blood level (0.4 m mol/L) required for therapeutic efficacy and the maximum level (1.5 m mol/L) beyond which toxicity may ensue.

The correct answer is: Lithium has a wide therapeutic range

Question **37**

Not answered

Marked out of 1.00

Flag question

A drug follows first order kinetics and has a t1/2 of 4 hours. In a healthy male how long will it take for this drug to reach steady state assuming no loading dose is given, and the drug is administered at equal intervals regularly?

Select one:

- 10 hours
- 2 hours
- 4 hours
- 20 hours

- 1 hour

It is estimated that it takes 4-5 t ½ for a drug to reach the steady plasma level.

The correct answer is: 20 hours

Question **38**

Not answered

Marked out of 1.00

Flag question

A 30-year-old is on combined oral contraceptive pill. The risk of becoming pregnant is increased by all except

Select one:

- Venlafaxine
- Topiramate
- St John's wort
- Carbamazepine
- Phenytoin

Explanation:

Psychiatric drugs that are CYP3A4 inducers (e.g. carbamazepine, St John's Wort, phenytoin, topiramate) reduce the bioavailability and hence contraceptive efficacy of combined hormonal contraceptives, the progesterone-only pill, and the subdermal implant both during use and for up to 28 days after stopping enzyme-inducer treatment. Thus, patients on enzyme inducers should be encouraged to use intrauterine contraception or injectables.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 314.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 855.

The correct answer is: Venlafaxine

Question **39**

Not answered

Marked out of 1.00

Flag question

Which of the following undergoes metabolism predominantly by first order kinetics?

Select one:

- Phenytoin
- Depot antipsychotics
- Venlafaxine XL
- Alcohol
- Thioridazine

Elimination of a constant quantity per time unit of the drug quantity present in the organism is called zero order kinetics. 95% of the drugs in use at therapeutic concentrations are eliminated by first-order elimination kinetics. A few substances are eliminated by zero-order elimination kinetics because their elimination process is saturated, especially at higher doses.

Examples are Ethanol, Phenytoin, high dose Salicylates, high dose Fluoxetine, high dose Omeprazole. Because in a saturated process the elimination rate is no longer proportional to the drug concentration but decreasing at higher concentrations, zero-order kinetics are also called 'non-linear kinetics' in clinical pharmacology. The absorption rate constant is slower than the elimination rate constant and, therefore, the depot antipsychotics exhibit 'flip-flop' kinetics where the time to steady-state is a function of the absorption rate, and the concentration at steady-state is a function of the elimination rate. (Elimination kinetics - UNIL, http://sepia.unil.ch/pharmacology/index.php?id=94 (accessed April 11, 2015).)

The correct answer is: Thioridazine

Question **40**

Not answered

Marked out of 1.00

Flag question

Drug B is a prodrug whose metabolite is an active hypnotic agent. If drug A induces the metabolism of drug B, which of the following is true?

Select one:

- Effects of drug B are more pronounced
- No change occurs in drug effect
- Effect of drug A will increase
- Hypnotic effect is lost
- Level of drug B's metabolite falls in blood

Prodrug's metabolism is increased. It will undergo more metabolism increasing the level of the active metabolite; drug B's effect will be more pronounced.

The correct answer is: Effects of drug B are more pronounced

Question **41**

Not answered

Marked out of 1.00

Flag question

Which of the following is an enzyme inducer?

Select one:

- Caffeine
- Smoking
- Grapefruit juice
- Paroxetine
- Valproate

Explanation:

Tobacco smoke contains polycyclic aromatic hydrocarbons that induce (increase the activity of) certain hepatic enzymes (CYP1A2 in particular). The extent of enzyme induction is determined by the number and type of cigarettes smoked and the degree of smoke inhalation. For some drugs used in psychiatry, smoking significantly reduces drug plasma levels and higher doses are required than in non-smokers. Smoking may also affect alcohol metabolism by inducing CYP2E1.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 856.

The correct answer is: Smoking

Question **42**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs has a metabolite that has the longest half-life?

Select one:

- Fluoxetine
- Duloxetine
- Venlafaxine
- Paroxetine
- Citalopram

Explanation:

Fluoxetine has a long half-life (2-3 days) and its active metabolite (norfluoxetine) an even longer half-life (2 weeks). Fluoxetine has a long half-life when compared with other SSRIs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 293.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 562, 742.

The correct answer is: Fluoxetine

Question **43**

Not answered

Marked out of 1.00

Flag question

The ratio between minimum plasma level causing toxic effect to minimum plasma level causing therapeutic effect is called

Select one:

- Toxic dose
- Therapeutic index
- Volume of distribution
- Affinity
- Potency

Explanation:

Some drugs have a narrow therapeutic index (a small increase in dose can cause toxicity and a small reduction in dose can cause a loss of therapeutic actions). The most commonly prescribed ones are digoxin, warfarin, theophylline, phenytoin, and lithium.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 602.

The correct answer is: Therapeutic index

Question **44**

Which one of the following is a potent hepatic enzyme inducer?

Not answered

Marked out of 1.00

Flag question

Select one:

- Gabapentin
- Carbamazepine
- Lamotrigine
- Valproate
- Lithium

Explanation:

Carbamazepine is a **potent inducer** of **hepatic cytochrome enzymes** and is metabolised by CYP3A4. Plasma levels of most antidepressants, most antipsychotics, benzodiazepines, warfarin, zolpidem, some cholinesterase inhibitors, methadone, thyroxine, theophylline, oestrogens, and other steroids may be **reduced by carbamazepine**. Valproate is hepatically metabolised; drugs that inhibit CYP enzymes can increase valproate levels (e.g. erythromycin, fluoxetine, and cimetidine).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 261, 266.

The correct answer is: Carbamazepine

Question **45**

Not answered

Marked out of 1.00

Flag question

Which one among the following is preferable to be used in the elderly than other tricyclic antidepressants?

Select one:

- Imipramine
- Dothiepin
- Trimipramine
- Amitriptyline
- Lofepramine

Explanation:

Lofepramine and nortriptyline are the least sedative TCAs. Lofepramine has moderate anticholinergic effects, although constipation/sweating can be severe. Postural hypotension can be a problem, but lofepramine is the least likely TCA to cause it. Lofepramine is relatively safe in overdose and is the TCA that is the least likely to cause cardiac conduction disturbance.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 421, 674.

The correct answer is: Lofepramine

Question **46**

Not answered

Which of the following types of drug molecules is most efficiently absorbed?

Select one:

Marked out of 1.00

Flag question

- Small, water soluble electrically neutral molecules
- Small, water soluble electrically charged molecules
- Large, water soluble electrically neutral molecules
- Small, lipid soluble electrically neutral molecules
- Large, lipid soluble electrically neutral molecules

To achieve a therapeutic effect most drugs need to cross at least one lipid membrane. This will depend on the size and shape of the molecule, lipid solubility and ionic charge. The blood-brain barrier only allows lipid-soluble molecules in the brain

The correct answer is: Small, lipid soluble electrically neutral molecules

Question **47**

Not answered

Marked out of 1.00

Flag question

The metabolism of which one among the following follow first order kinetics?

Select one:

- Alcohol
- Oral Antipsychotic
- Depot antipsychotic
- Absorption of controlled release drugs
- Phenytoin

Most drugs show first order kinetics where the rate of absorption or elimination is directly proportional to the amount of drug remaining. The concentration of a drug in the plasma will normally follow first order kinetics, decreasing in an exponential manner related to plasma concentration. However, drugs like ethanol and Phenytoin are metabolised by zero order kinetics, their concentration decreasing in a linear fashion i.e., at a constant rate irrespective of plasma concentration. With zero-order kinetics, a fixed amount of drug is absorbed each time independent of drug concentrations and hence a small increase in drug ingestion results in a large increase in plasma concentration. Absorption of controlled release drugs and depot antipsychotics also follow zero-order kinetics.

The correct answer is: Oral Antipsychotic

Question **48**

Not answered

Marked out of 1.00

Flag question

Which of the following medications has the least protein-binding?

Select one:

- venlafaxine
- paroxetine
- valproate
- clomipramine
- fluoxetine

Explanation:

Venlafaxine, a phenylethylamine, is a relatively weak serotonin (5-HT) and weaker norepinephrine (NE) reuptake inhibitor with a 30-fold difference in binding of the two transporters. It has a clear dose progression, with low doses predominantly binding to the 5-HT transporter and more binding of the NE transporter as the dose ascends. Venlafaxine is metabolised to the active metabolite O-desmethylvenlafaxine (ODV; desvenlafaxine) by CYP2D6. Both venlafaxine and ODV have low protein binding (27% and 30% respectively).

Ref: Shelton RC. Serotonin and norepinephrine reuptake inhibitors. In Antidepressants 2018 (pp. 145-180). Springer, Cham.

The correct answer is: venlafaxine

Question **49**

Not answered

Marked out of 1.00

Flag question

There is poor oral absorption of most psychotropics in which of the following parts of the gastrointestinal system?

Select one:

- Ileum
- Stomach
- Jejunum
- Colon
- Rectum

Explanation:

The GI tract begins at the mouth, includes the esophagus, stomach, small intestine, large intestine (also known as the colon), and rectum, and ends at the anus. Generally, absorption from the stomach is minor compared to the intestine because of the gastric epithelial barrier preventing diffusion, the relatively small available surface area, and the negligible expression of nutrient (uptake) transport proteins. The small intestine (duodenum, jejunum, and upper region of ileum) is the major and most efficient site for absorption of most drugs. The colonic absorption of drugs can differ significantly from absorption in the small intestine as a consequence of several physiological, physiochemical, and biopharmaceutical factors. Interest has developed in directing drugs and dosage forms to effect primary drug release to the colon. Targeted distribution of an orally administered drug at the large intestine confers therapeutic treatment of colonic diseases, peptide and protein therapy, and chronotherapy.

Ref: Yang Y et al. Oral drug absorption: Evaluation and prediction. In Developing solid oral dosage forms 2017 Jan 1 (pp. 331-354). Academic Press.

The correct answer is: Stomach

Question **50**

Not answered

Marked out of 1.00

Flag question

Which of the following enzymes are responsible for oxidative metabolism of most psychotropic drugs?

Select one:

- Cytochrome P450 family
- Glycoprotein P
- Aryl Sulfatase

- Monoamine Oxidase
- G Protein

Explanation:

The cytochrome P450 (CYP450) enzyme system mediates how the body metabolises many drugs, including antipsychotics. The CYP450 enzyme in the gut wall or liver converts the drug into a biotransformed product in the bloodstream. After passing through the gut wall and liver, the drug will exist partly as unchanged drug and partly as biotransformed drug.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 49.

The correct answer is: Cytochrome P450 family

Question **51**

Not answered

Marked out of 1.00

Flag question

Valproate levels are decreased by

Select one:

- Lithium
- Cimetidine
- Erythromycin
- Carbamazepine
- Fluoxetine

Explanation:

Carbamazepine is a **potent inducer** of **hepatic cytochrome enzymes** and is metabolised by CYP3A4. Plasma levels of most antidepressants, most antipsychotics, benzodiazepines, warfarin, zolpidem, some cholinesterase inhibitors, methadone, thyroxine, theophylline, oestrogens, and other steroids, as well as valproate, may be **reduced by carbamazepine**.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 266.

Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. p. 832.

The correct answer is: Carbamazepine

Question **52**

Not answered

Marked out of 1.00

Flag question

The enhancement of drug effects following the repeated administration of the same dose of a drug is called

Select one:

- Tolerance
- Neuroadaptation
- Habituation
- Sensitisation

- Cross Tolerance

Tolerance refers to the diminished response to the administration of a drug after repeated exposure. It may be behavioural tolerance through learning to cope with the effects, increased metabolism, reduced receptor sensitivity or number, activation of a homeostatic mechanism. Cross-tolerance refers to shared tolerance among drugs with similar chemical structure. Cross-tolerance forms the basis for a number of drug interactions (E.g. alcohol with barbiturates). Sensitisation refers to the enhancement of drug effects following the repeated administration of the same dose of the drug. Drug sensitization occurs in drug addiction and is the physiological opposite of drug tolerance.

The correct answer is: Sensitisation

Question **53**

Not answered

Marked out of 1.00

Flag question

Which one among the following statements is true with respect to the interaction of fluoxetine with other drugs?

Select one:

- Fluoxetine reduces the effects of propranalol
- Fluoxetine reduces the plasma concentration of carbamazepine
- Fluoxetine reduces the concentration of haloperidol
- Fluoxetine enhances the effects of oral anticoagulants
- Fluoxetine reduces the plasma concentration of phenytoin

Explanation:

Fluoxetine inhibits CYP450 2D6 and 3A4. Fluoxetine may displace high protein-bound drugs (e.g. warfarin). There is a possible risk of increased bleeding with fluoxetine, especially when combined with anticoagulants (eg. warfarin, NSAIDs). Via CYP 450 2D6 inhibition, fluoxetine can increase the plasma levels of some beta blockers. Fluoxetine increases concentration of haloperidol, carbamazepine and phenytoin.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Prescriber's Guide, 7th Edition. 2021. pp. 303-304.

The correct answer is: Fluoxetine enhances the effects of oral anticoagulants

Question **54**

Not answered

Marked out of 1.00

Flag question

Which of the following is true concerning tricyclic metabolism?

Select one:

- Elderly clear more tricyclics than adults
- Adolescents clear more tricyclics than adults
- None of the drugs need plasma monitoring
- Metabolism is not affected by CYP enzyme polymorphisms
- They do not interact with chlorpromazine

Tricyclic antidepressants are not effective in prepubertal children but may have marginal efficacy in adolescents with depression. Due to more extensive metabolism, young people require higher mg/kg doses than adults.

The correct answer is: Adolescents clear more tricyclics than adults

Question **55**

Not answered

Marked out of 1.00

Flag question

What is the volume of distribution of drug A if the plasma concentration immediately after IV administration of 10mg is 0.01mg/mL?

Select one:

- 100mL
- 1000mL
- 1 mL
- 10mL
- 0.1mL

(Volume of distribution) Vd =Q/Cp, where Vd-volume of distribution, Q-quantity of drug and Cp-plasma concentration at time of administration ('zero time').

The correct answer is: 1000mL

Question **56**

Not answered

Marked out of 1.00

Flag question

Which of the following should be considered when switching from one brand of medication to another brand?

Select one:

- Rate of absorption
- Bioequivalence
- Bioavailability
- Volume of distribution
- Clearance

Bioequivalence is important feature to be considered when changing from one brand to other brand of the same compound e.g. camcolit vs. priadel for lithium carbonate or Clozaril vs. zaponex for clozapine. It is a measure of comparability of plasma levels of two different formulations of the same active compound when given at same dose and same route of administration

The correct answer is: Bioequivalence

Question **57**

Not answered

Marked out of 1.00

Most orally administered psychotropic drugs undergo hepatic metabolism followed by renal clearance. Which of the following statement about clearance is true?

Select one:

Flag question

- Clearance can occur only via urine
- Clearance rate determines half life of a drug in the body
- It is fixed for each individual with minor variations among drugs
- It is the amount of drug eliminated from the body in a given time
- It always depends on the dose consumed

Clearance is defined as the volume of blood cleared of a particular drug in unit time. It is specific for each drug and does not depend on drug concentration in plasma (because if concentration increases, elimination will also increase under first order kinetics). It represents the relationship between the rate of drug elimination (t1/2) and plasma level. For drugs with first order kinetics, clearance is constant irrespective of dose consumed because the rate of elimination is directly proportional to plasma level. Total body clearance depends on renal and non-renal clearance such as sweat, bile etc.Renal elimination without significant liver breakdown is seen in lithium, amisulpride, sulpiride, gabapentin, acamprosate and amantadine.

The correct answer is: Clearance rate determines half life of a drug in the body

Question **58**

Not answered

Marked out of 1.00

Flag question

Which of the following atypical agents have the shortest half life?

Select one:

- Quetiapine
- Aripiprazole
- Olanzapine
- Clozapine
- Risperidone

Explanation:

Quetiapine has a short half-life (about 7 hours), and optimal dosing is two or three times per day.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 611.

The correct answer is: Quetiapine

Question **59**

Not answered

Marked out of 1.00

Flag question

The SSRI with longest total half-life is

Select one:

- Citalopram
- Fluoxetine
- Duloxetine
- Sertraline
- Paroxetine

Explanation:

Fluoxetine has a long half-life (2-3 days) and its active metabolite (norfluoxetine) an even longer half-life (2 weeks). Fluoxetine has a long half-life when compared with other SSRIs.

Ref: Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 293.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. pp. 562, 742.

The correct answer is: Fluoxetine

Question **60**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs have a half-life of 3 days?

Select one:

- Quetiapine
- Reboxetine
- Aripiprazole
- Carbamazepine
- Fluoxetine

Explanation:

The mean elimination half-life of aripiprazole is about 75 hours. It has a weakly active metabolite with a half-life of 96 hours. These relatively long half-lives make aripiprazole suitable for once-daily dosing.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 612.

The correct answer is: Aripiprazole

Question **61**

Not answered

Marked out of 1.00

Flag question

A newly developed antipsychotic is said to be highly permeable across lipid membranes and so achieve very high bioavailability when given orally. Which of the following properties of the drug enable this?

Select one:

- Balanced hydrophilic and lipophilic properties
- Low concentration gradient across membranes
- Low protein binding
- Exists in highly ionised form predominantly
- All of the listed options

If highly hydrophilic; the drug cannot cross the lipid cell membrane. On the other hand, if

highly lipophilic, the drug is not soluble enough to cross the water layer adjacent to the cell.Balanced hydrophilic and lipophilic property is essential for high permeability.

The correct answer is: Balanced hydrophilic and lipophilic properties

Question **62**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs has the least propensity to cause a withdrawal effect?

Select one:

- Temazepam
- Zolpidem
- Buspirone
- Zopiclone
- Zaleplon

Explanation:

Buspirone does not cause discontinuation/withdrawal symptoms. Because it does not act on the GABA-chloride ion channel complex, it is not recommended for the treatment of withdrawal from benzodiazepines, alcohol, or sedative-hypnotic drugs, except as treatment of comorbid anxiety symptoms.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 673.

The correct answer is: Buspirone

Question **63**

Not answered

Marked out of 1.00

Flag question

Zaleplon has a half-life of approximately

Select one:

- 12-24 hours
- More than 24 hours
- 2-3 Hours
- 1 hour
- 3-6 Hours

Explanation:

Zaleplon reaches peak plasma concentrations in 1 hour and has a half-life of 1 hour.

Ref: Boland R, Verduin M. Kaplan and Sadock's Synopsis of Psychiatry, 12th Edition. 2022. p. 679.

The correct answer is: 1 hour

Question **64**

Not answered

Which one among the following statements is incorrect about the oral route of administration?

Marked out of 1.00

Flag question

Select one:

- Preparations that are soluble in gastrointestinal fluids are absorbed better
- Oral route drugs are subject to renal first pass metabolism
- Lipid solubility aids absorption
- Oral route leads to variable plasma concentration because the absorption may be erratic
- Acid-resistant drugs are better absorbed

Drugs taken orally are subject to metabolism by the liver (First-Pass Effect). A drug taken orally is absorbed from the gut into the enterohepatic circulation, where the liver inactivates some of the drug before even it reaches the systemic circulation, which is known as the first pass effect. It can be avoided if the drug is injected intravenously whereby the drug directly enters the systemic circulation.

The correct answer is: Oral route drugs are subject to renal first pass metabolism

Question **65**

Not answered

Marked out of 1.00

Flag question

A patient suddenly withdraws from a medication. How many half-lives does it take for the medication to be completely excreted?

Select one:

- 5
- 3
- 2
- 8
- 10

From a clinical standpoint, it is common to assume that a drug is effectively eliminated after 4-5 half-lives, assuming first order kinetics

The correct answer is: 5

Question **66**

Not answered

Marked out of 1.00

Flag question

Urinary acidification can help eliminate which of the following medications on overdose?

Select one:

- Amphetamines
- Salicylates
- Paracetamol
- Tricyclics
- Barbiturates

Explanation:

Amphetamine intoxication can be treated with ammonium chloride. The recommended dose to acidify the urine is ammonium chloride 500mg every 2-3 hours.

Ref: https://emedicine.medscape.com/article/289973-treatment

The correct answer is: Amphetamines

Question **67**

Not answered

Marked out of 1.00

Flag question

How many minimum weeks of interval should be given between fluoxetine and phenelzine when switching from the SSRI to the MAOI?

Select one:

- 5 weeks
- 1 week
- 12 weeks
- 3 weeks
- 2 weeks

Explanation:

When switching from fluoxetine to phenelzine, it is advised to stop fluoxetine then wait 5-6 weeks before starting phenelzine.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 338.

The correct answer is: 5 weeks

Question **68**

Not answered

Marked out of 1.00

Flag question

A new molecule investigated for anxiety disorder failed at initial stages of animal testing as it failed to reach the brain crossing the blood-brain barrier. Which of the following factors decide the degree of blood-brain barrier permeability?

Select one:

- Frequency of administration of drug
- Time of drug administration
- Half life of drug
- Lipid water partition coefficient
- Nitrogen content of the drug

The ability of a drug to pass blood brain barrier depends on its molecular size, lipid solubility and ionic status. Unionized molecules that are freely available and less protein bound are transported across the barrier easily. In general higher the lipid-water partition coefficient, greater the ability to cross the barrier

The correct answer is: Lipid water partition coefficient

Question

69

Not answered

Marked out of 1.00

Flag question

Which of the following factors can improve absorption of orally administered drugs?

Select one:

- High hydrophilic capacity of drug
- Fast intestinal motility
- Presence of P glycoprotein
- Increasing particle size of the administered drug
- None of the listed options

Explanation:

This is mainly due to lack of absorption from the intestine related to the presence of inhibitory factors like food, or gastric acid or due to changes in intestinal motility (e.g. having diarrhoea or vomiting can affect drug absorption). Inherent properties of certain drugs can also affect their absorption (e.g. highly hydrophilic drugs cannot cross the lipid cell membrane, while highly lipophilic drugs will struggle to cross the water layer in extracellular space). Inhibition of P-glycoprotein (e.g. by grapefruit juice) can increase absorption of certain medications. Reducing particle size increases surface area and can increase dissolution and thus absorption.

Hua S. Advances in oral drug delivery for regional targeting in the gastrointestinal tract-influence of physiological, pathophysiological and pharmaceutical factors. Frontiers in pharmacology. 2020 Apr 28;11:524.

The correct answer is: None of the listed options

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